

Customer Shopping Behavior Analysis

Data Analytics Project Report

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1. Executive Summary

This report presents an end-to-end data analysis of customer shopping behavior, covering 3,900 transactions across four product categories: Clothing, Accessories, Footwear, and Outerwear. The project followed a complete analytics workflow — data cleaning and feature engineering in Python (Pandas), database loading via SQLAlchemy into PostgreSQL, business analysis using SQL, and visualization through an interactive Power BI dashboard.

The analysis addressed five core business questions covering revenue by gender, discount-driven spending behavior, top-rated products, shipping type comparison, and the impact of subscription status on customer spend. Key findings are summarized in Section 4.

2. Objectives

- Clean and prepare raw transactional data for reliable analysis.
- Engineer new features (age groups, purchase frequency in days) to enrich analytical depth.
- Answer specific business questions using SQL to support data-driven decision-making.
- Visualize findings in an interactive Power BI dashboard for stakeholder-friendly exploration.

3. Methodology

3.1 Data Cleaning & Preprocessing (Python)

- Filled missing Review Rating values using the median rating per product category.
- Standardized column names to lowercase, snake_case format.
- Engineered an age_group feature (Young Adult, Adult, Middle-aged, Senior) using quartile-based binning.
- Engineered a purchase_frequency_days feature, mapping categorical frequency labels (e.g. Weekly, Monthly) to numeric day intervals.
- Identified and removed the promo_code_used column after confirming it was identical to discount_applied across all records.
- Loaded the cleaned dataset into a PostgreSQL database using SQLAlchemy for downstream SQL analysis.

3.2 SQL Analysis

Five targeted queries were written in PostgreSQL syntax to answer the project's core business questions, using aggregation, grouping, and subqueries. Full query code is available in the project repository.

3.3 Visualization (Power BI)

Findings were consolidated into an interactive Power BI dashboard, enabling filtering and exploration of revenue, ratings, shipping, and subscription patterns by category and demographic.

4. Key Findings

4.1 Revenue by Gender

Gender	Total Revenue (USD)	Customers
Male	\$157,890	2,652
Female	\$75,191	1,248

Male customers generated roughly 2.1x the total revenue of female customers. This gap is driven largely by transaction volume — male customers account for 68% of all transactions in the dataset — rather than a higher average spend per order.

4.2 Discount Usage vs. Above-Average Spend

Out of all customers who applied a discount, 839 still spent more than the overall average purchase amount of \$59.76. This indicates that a meaningful segment of price-sensitive shoppers are also high-value customers — discount usage alone does not predict lower order value.

4.3 Top 5 Highest-Rated Products

Rank	Product	Avg. Rating (out of 5)
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.79

Accessory and footwear items dominate the top-rated list, suggesting stronger customer satisfaction in these categories compared to the dataset's overall average rating of 3.75.

4.4 Standard vs. Express Shipping

Shipping Type	Avg. Purchase Amount (USD)
Express	\$60.48
Standard	\$58.46

Customers who chose Express shipping spent, on average, about \$2 more per order than those who chose Standard shipping — a modest but consistent difference suggesting a slight willingness to pay more among faster-shipping customers.

4.5 Subscription Status Impact

Subscription	Customers	Avg. Spend (USD)	Total Revenue (USD)
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

Average spend per order is nearly identical between subscribed and non-subscribed customers (\$59.49 vs. \$59.87), indicating that subscription status does not meaningfully influence order value. Non-subscribers contribute far more total revenue, but this reflects their larger share of the customer base (73%) rather than higher individual spending.

5. Conclusion

This project demonstrates a complete analytics workflow from raw data to actionable business insight. The findings suggest that revenue is driven more by customer volume and category mix than by shipping method or subscription status alone, while discount usage does not necessarily indicate lower-value customers. These insights could inform targeted marketing strategies, particularly around underrepresented customer segments and top-rated product categories.

6. Tools & Technologies

Tool	Purpose
Python (Pandas)	Data cleaning, preprocessing, feature engineering
SQLAlchemy + PostgreSQL	Loading cleaned data into a relational database
SQL	Business question analysis via queries
Power BI	Interactive dashboard and visualization

7. Project Links

GitHub Repository: github.com/AbdullahHasan707/Customer-Shopping-Behavior-Analysis

LinkedIn: [linkedin.com/in/abdullah-hasan-shah-b306b6370](https://www.linkedin.com/in/abdullah-hasan-shah-b306b6370)